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Improved prediction of drug-drug interactions using ensemble deep neural networks

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ABSTRACT

Nowadays, combining multiple drugs is the optimal therapy to decelerate the pathologic process, which contains various underlying adverse effects due to drug-drug interactions (DDIs). Artificial intelligence (AI) has the potential to evaluate the interaction, pharmacodynamics, and possible side effects between drugs. Over the past years, many AI-based DDI prediction techniques, including both machine learning and deep learning, that harness available big data have been presented. Although significant progressives have been gained through previous methods, improvements are still crucial. In this work, we introduce an ensemble deep neural network that can help to improve the predictive performance of DDIs. As a result, our prediction model could predict 86 types of DDIs on a benchmark dataset with an average accuracy of 93.80%. Our ensemble classifier produces better performance when compared with the existing proposed methods on the same dataset. Our model's high performance places it among the top list of those well-built pharmacovigilance-assisted tools that facilitate the detection of DDIs to support medical decisions and drug development. We released our source codes and models at https://github.com/khanhlee/edn-ddi.

1. Introduction

The event of a drug's pharmacological effect being modified by that of another drug when administered in a combined regimen is known as drug-drug interactions (DDIs). Studies have shown that the percentage of polypharmacy instances is rapidly expanding its realm, especially in older people, with almost 30 % of those among the age group of 65 and higher in developed countries take above four drugs at a time [1]; and 67 % of American elders were on a five-or-more combined medications in 2010–2011 [2]. As a linear force of increasing DDIs, polypharmacy is often associated with adverse drug events (ADEs) that might bring more harm than good to those on multiple drug therapies. Research on DDIs in hospitalized patients by Zheng shows that almost two-thirds of patients receiving critical medical care develop at least one possible DDI while staying in the hospital [3]. Many patients visit the emergency sections in the hospital due to unwanted drug interactions that cause adverse drug events, usually leading to more severe outcomes [4]. The estimated number of those on multiple drug therapies who suffered from these events ranged between 3 and 5 % among all medication errors in the hospital. Also, emergency visits from outpatients due to adverse drug events are considered a significant cause of morbidity in the United States [5]. However, while it is inevitable to avoid polypharmacy for the treatment of various diseases or the synergistic effects in pharmacological therapy, a better understanding of DDIs, especially the antagonistic or unwanted DDIs, can significantly contribute to managing and designing the best achievable pharmacology therapy for patients. Undeniably, verifying DDIs through wet-lab experiments can exhaust researchers' time and resources. On top of this, high throughput screening in-vivo models is impossible when testing many drug combinations. To solve this problem, many drug databases have recently been continuously constructed and updated to smooth the way for

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the application of computational prediction models in DDIs determination to lower time and cost [6,7].

Over the past years, as the drug database rapidly perfects itself, many computer-based DDI prediction techniques, including both machine learning-based and deep-learning-based techniques, that harness available big data have been presented [8]. Earlier in 2013, Vilar et al. used interaction profile fingerprints as a measurement method to evaluate the similarity of drugs as pairs and generated new DDIs for other pairs with non-cross interactions [9]. However, this database-lean-on method only used the known DDIs while overlooking other drug data. Later, the similarity-based method was applied when assuming that similar drugs might interact with the same drugs. Vilar et al. have applied this method to predict new DDIs using structural and interaction profile fingerprint similarity [10]. On the same similarity-based trend, Cheng et al. built a support vector machine model that takes input, including the features extracted from the simplified molecular-input line-entry system (SMILES) information coupled with the side effect similarity data [11]. Furthermore, different feature extraction methods have also been discovered to improve the predictive performance; Park et al. predicted pharmacodynamic DDIs through protein interaction signaling propagation using a random walk restart algorithm [12].

Along with the advancement in deep learning techniques that can independently execute feature engineering, a significant improvement in deep learning-based DDIs prediction is within expectations. Deng et al. proposed a multimodal deep learning framework to predict 65 categories of DDI events [13]. Rohani et al. suggested a new neural network architecture that utilizes integrated similarity to enhance performance [14]. Subsequently, different types of neural networks have been assessed for this DDI prediction, such as autoencoders and deep feedforward networks [15], artificial neural networks and factor propagation [16], graph convolutional neural network (GCNN), and bidirectional long short-term memory (BiLSTM) recurrent neural networks [17]. Specifically, Ryu et al. used structural information and names of paired drugs as input and provided benchmark data containing 86 DDIs. Their deep learning model reached a performance of 92.4 % [18].

Although significant progressives have been gained through previous methods, improving the prediction accuracy is still crucial. In this work, we introduce an ensemble neural network for predicting DDIs based on the given data set provided in Ryu's research [18]. Our ensemble-based classifier produces better performance when compared to the existing proposed method of Ryu et al. [18]. The performance improvement achieved by our model has been proven significant using standard metrics for predictive model evaluation. In detail, we have considered three modalities of modeling methods, including deep neural network (DNN), random forest (RF), and eXtreme Gradient Boosting (XGBoost), to analyze the effect of the ensemble deep learning model in predicting DDIs. All three models are optimized using the RandomizedSearchCV method to find the best hyper-parameter and cross-validate to avoid overfitting. Even though our model uses the previous method that takes the structural similarity profile and name of drug pairs as input, our ensemble deep neural network architecture (EDN) is highly tuned for this task. Our EDN model represents a reliable framework for accurate DDIs prediction.

2. Materials and methods

2.1. Experimental design

Fig. 1 shows a scheme of our workflow, which included the following steps: (1) Extracting the cheminformatics data of drugs; (2) Portraying the numerical representation of a drug pair as a structural similarity profile-based feature vector (SSP); (3) Conducting preliminary selection for suitable sub-models within the expected performance range; and (4) Building EDN model base on the ensemble of those selected sub-models.

2.2. Benchmark dataset

To analyze the efficacy of our study, we employ the benchmark dataset proposed by Ryu et al. [18] that covers 86 DDIs types classified from the gold standard dataset holding 192,284 DDIs contributed by 191,878 drug pairs in DrugBank [7]. This dataset contained the molecular form of each molecule used, and we did not apply any preprocessing. Fig. 2 shows the distribution of the data among 86 DDIs types.

Regarding our study's structural data resources for *in silico* DDI extractions and prediction, we decided to utilize the cheminformatics data of individual drugs provided by Ryu et al. [18] as the starting point to prepare our dataset. The cheminformatics information of the two drugs was then combined and related to the fixed target for collation in the proven set of DDIs found in Drug Bank.

2.3. Chemical features and technical processing

Our EDN model manipulates the molecular structural information of drugs to do the DDIs prediction task. The quickly learned and workable notation, SMILES, can be considered an optimal choice to represent a drug's chemical structure that enables the computer to understand and handle it [19]. Thereby, it allows the cheminformatics data of the drug to be extracted easily. Through this cheminformatics data convey mixed and sufficient details, drug information can be transformed into valuable knowledge that can contribute remarkably to DDIs prediction problems.

Regarding extracting drug cheminformatics information to associate its feature with the DDIs types set in our work, we realized that using a feature-vector-based delineating structural characteristics of individual drugs is a common and practical approach. In our study, we retrieved the drug cheminformatics data through its SSP. SSP contains pairwise structural similarity scores obtained from the comparison between the input drug and all the 2,159 approved drugs of DrugBank as a fixed comparison target. We took the evaluation method between a drug pair by using the Tanimoto coefficient [20], measuring between that pair's binary fingerprints extracted from their SMILES [21].

Over the past decade, there have been two standard methods to generate this numerical representation of the drug: Functional-Class Fingerprint (FCFPs) and Extended Connectivity Finger Printing (ECFPs) methods. After considering both options, we decided the ECFPs method would work best for our goal because of its specificity in structure–activity modeling [21]. ECFPs were calculated using a Python package *RDKit* (version 2013.09.1; https://www.rdkit.org), and a diameter of 4 was selected (ECFP4). Although some latest versions of RDKit have been released in several years, we kept using the version 2013.09.1 to make our study consistent with the previous predictor [18]. As accompanied, our model input variables were all generated via this method. These circular topological fingerprints portray the drug's molecular structures under circular atom neighborhoods. The pivotal point of using this method is that it is designed to heed both the presence and absence of functionality, which are all vital elements in molecular activity analysis. The final folded list of identifiers is a 1024/2048 bit-vector. To avoid the curse of dimensionality that refers to unexpected prediction outcomes, the dimension of SSPs was reduced by implementing principal component analysis (PCA) using a Python package *scikit-learn* (https://scikit-learn.org/). For this, a dimension of 50 per molecule was used as it showed the most remarkable prediction accuracy. Thus a given molecular pair had 100 components and was used as input to our models.

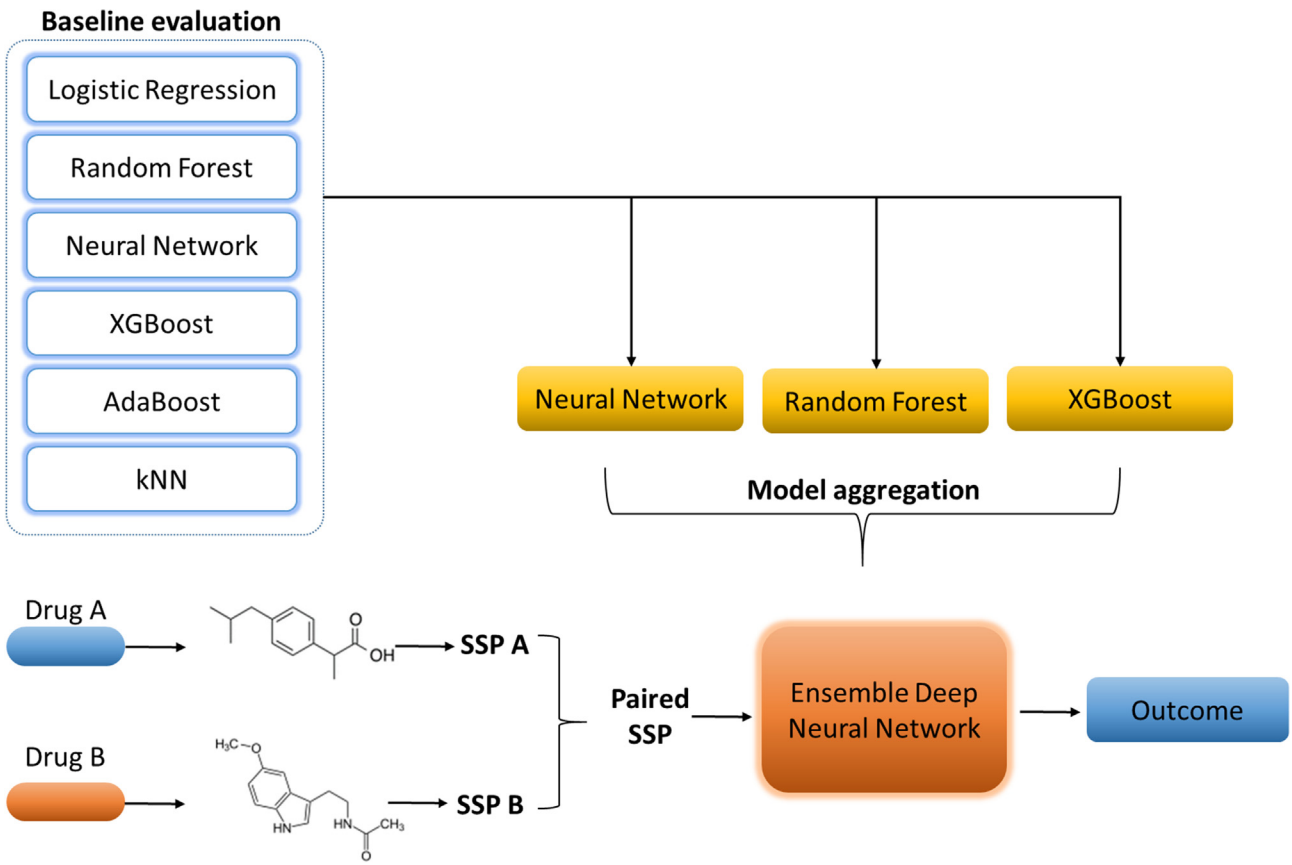

Fig. 1. Ensemble deep neural network workflow.

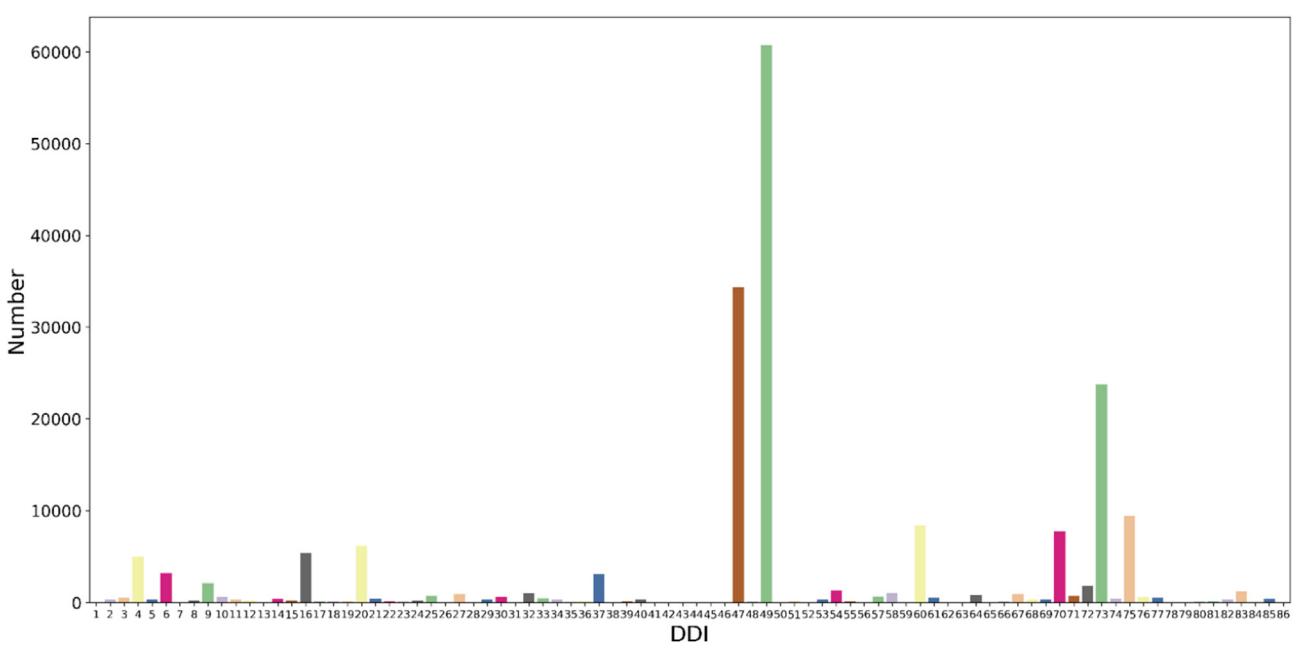

Fig. 2. Data distribution in the dataset of 86 DDIs types.

2.4. EDN overview

In this study, we first built six machine learning models using the pre-processed dataset (as shown in Fig. 1) to create an ensemble model. The algorithms used were Logistic Regression (LR), RF, DNN, XGBoost, k-nearest neighbor (kNN), and Adaptive Boosting (AB). Each algorithm has its unique characteristics and advantages. For example, RF, XGBoost, and AB algorithms are based on decision trees, although the number of trees generated and the method for model fitting are different. Subsequently, our EDN model was built from a combined set of

three preliminary filtered classifiers via the accuracy of predictive performance. These include DNN, RF, and XGBoost. All algorithms were implemented using Python programming langue via *scikit-learn* package (machine learning) and *tensorflow* (deep learning). We also applied a cross-validation grid search to all algorithms to find out the optimal hyperparameters of each one.

2.5. Statistical analysis

To evaluate our model's performance, we applied 5-fold cross-validation tests to confirm unbiased evaluation. This study exploited the evaluation criteria based on commonly used metrics for classification problems, including precision, recall, and accuracy [22].

Precision (positive predictive value) indicates the rate of correct predictions among all predicted interactions and is defined as follows:

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

Recall (sensitivity) indicates the ratio of actual positive interactions that were predicted as positive.

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative}$$

Accuracy indicates the total accuracy of positive and negative interactions.

$$Accuracy = \frac{TruePositive + TrueNegative}{AllSamples}$$

3. Results and discussions

3.1. Data acquisition and processing

Since the individual drugs' SSP were dimensionally reduced to 50 variables to avoid the curse of dimensionality, no further transformation is needed. The dataset includes no personal information and is publically available. As the first step, we combined the cheminformatics data of two drugs and related it to the proven data set in DrugBank to calculate pairwise structural similarity scores for SSP. As a result, a total of 100 variables were used for each drug pair. The final input data set contain 20 million cells, 105 columns, and 191 thousand rows. The data exploration for the final dataset is then finished with no missing value observed. Worthy of paying attention, the target variable is imbalanced; most DDIs occur at types 47th, 49th, and 73rd. However, since we are working on the medical data, no data manipulation method should be applied to balance the dataset. Besides, the PCAs of all drug pairs are well distributed except for three variables: PCA1, PCA3, and PCA93. The detail of the benchmark data can be found in the previous paper [18] and Table 1.

3.2. Analysis and modelling

According to the 5-fold cross-validation results of the individual models, three outperforming models were then selected, and an ensemble model was constructed using three selected models. Fig. 3 illustrates the performance of different single models in our preliminary filtration round to find suitable candidates for our ensemble method. Therefore, we constructed the ensemble model based on DNN, RF, and XGBoost (the base models outweighed other classifiers in our preliminary accuracy estimation), to boost the DDIs prediction performance (Fig. 3). The final output of the ensemble model was the stacking result of the individual outputs using another LR classifier.

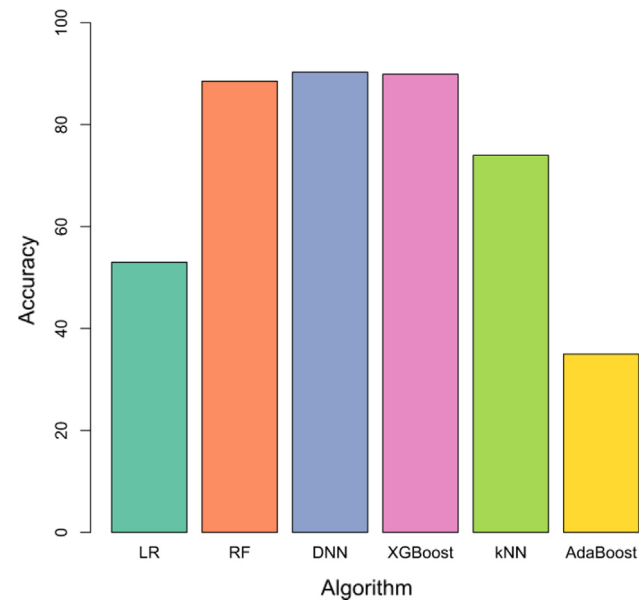

Fig. 3. Comparison among different machine learning or deep learning algorithms. Random Forest (RF), deep learning (DL), XGBoost (XGB) outperform other candidates.

3.3. Validation of results and comparison to State-Of-The-Art study

Some previous studies have implemented DDI predictors using different methods such as sparse feature learning ensemble method [23], multimodal deep learning [13], substructure–substructure interactions [24], and transformer self-attention mechanism [25]. However, only the benchmark study of Ryu et al. [18] used the same dataset as us, thus we compare the performance of our EDN model to this study to ensure a fair comparison. In Ryu's method, they manipulated the structural information of drugs to make DDIs predictions. The model used SSP to measure the similarity of drug pairs getting from the DrugBank and applied the DNN with optimized architecture to deal with this problem. Table 2 describes the performance evaluation of all methods and the baseline. Our EDN model achieves a surprising accuracy score of 93.80 % compared to the performance of individual sub-models and the baseline study. Other metrics have also shown a significant performance of our model compared to the benchmark study (average recall of 94.29 % and average precision of 83.78 %). This performance also means that our model was superior to another predictor namely HNAI [11] for the same prediction purpose. The precision and recall results of EDN across 86 DDIs types is illustrated in Fig. 4. The overall results demonstrated that EDN achieved better performances compared to other methods (benchmark paper and individual machine learning models).

Table 1
Details of benchmark dataset.

Data Source	DDIs	Drug pairs	DDI types	Similarity type
DrugBank (version 5.0.3)	192,284	191,878	86	Chemical structure

Table 2
Performance comparison of different methods with the same data set.

Method	Algorithm	Testing Accuracy
Ryu et al.	Deep neural network	92.4
Sub-model 1	Deep neural network	90.33
Sub-model 2	Random Forest	88.53
Sub-model 3	XGBoost	89.81
Ensemble	Ensemble deep neural networks	93.80

3.4. Experimental discussions

In this paper, we report our EDN model capable of transforming the chemical structures of drugs and their names into useful predictive information that can contribute to critical decision-making in drug prescription and the development of new medicines. EDN takes on an optimized ensemble model of DNN, RF, and XGBoost together with the drug feature vector (SSP), which has shown superior accuracy (93.80 %) compared to the analogous classifier in previous research [18]. Several reasons explain to the high performance of EDN can be:

- It exploits the critical information of a drug pair, the structural information that represents deep insight about them via the similarity-based method.
- It has been highly tuned for DDIs prediction by our customizing architecture, applying the RandomSearchCV method, and controlling overfitting by cross-validation.
- The neural network-based model can automatically process and extract high-level features that can yield better performance [26].

The performance of EDN has been evaluated via two mechanisms: first, the statistical significance of model performance has been tested

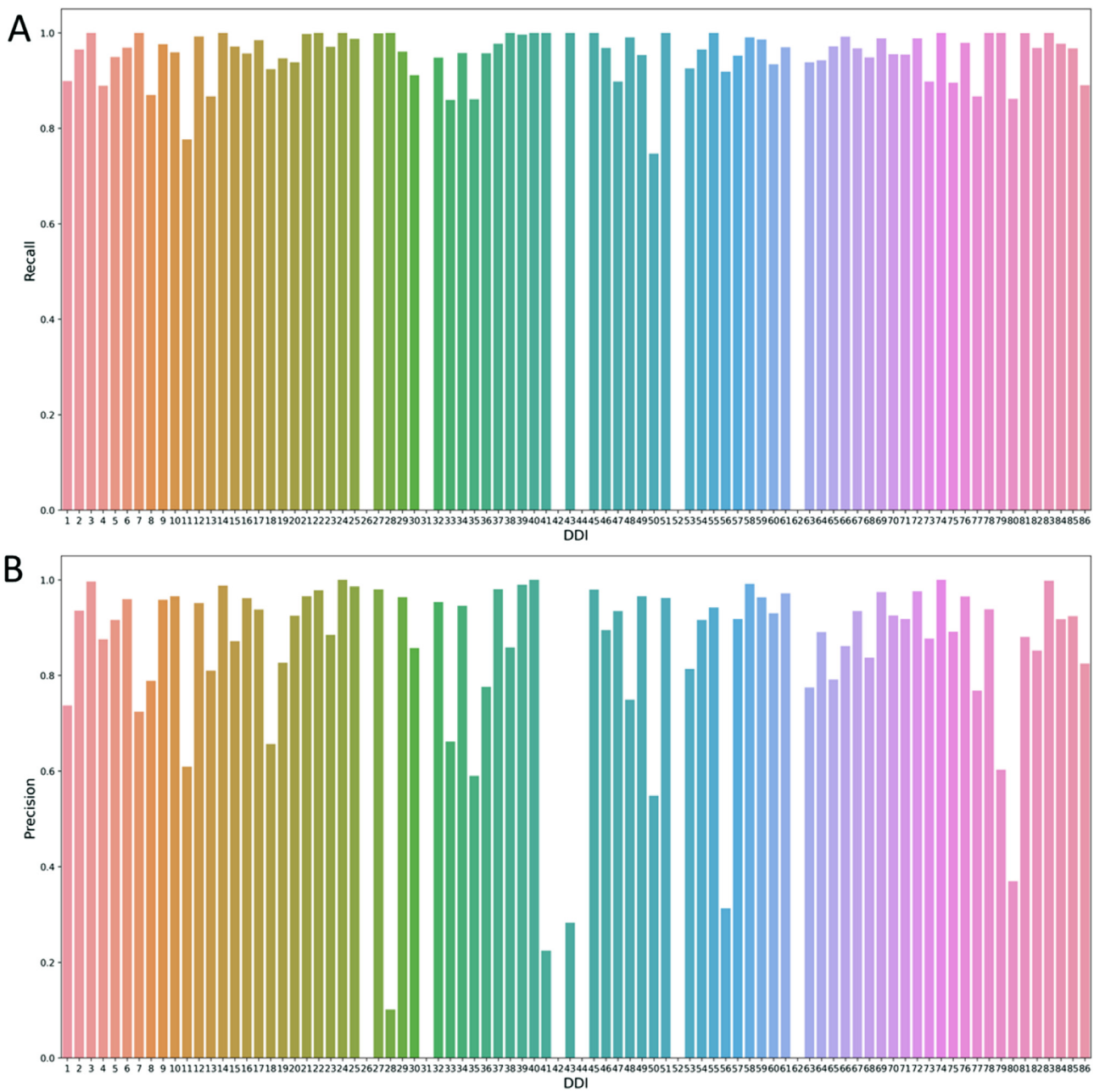

Fig. 4. Recall and precision results across 86 DDIs types. (A) Recall, (B) Precision.

regarding the evaluation scores for predictive models. Second, the accuracy of EDN prediction model has been collated with baselines research results [18] and outperformed their model performance. It should also notice that the neural network-based method requires powerful hardware, and the computer running it will have to shoulder the burden of memory usage. In overcoming this challenge, we apply the numeric values representing format conversion to aim for the suitable type that can be most capable of saving our memory usage.

Previous studies have made use of many other similarity matrices to fully extract different information about drugs outside of only the structural information [11,27,28]. However, this may have to be exchanged for a much longer time of training the model. In more detail, the less complex the similarity metrics are for the model to learn and train, the faster it will link up, the less noise will be seen, and the more likely it is that a mathematically optimal time for model training will be found. In addition to this, another limitation that is worthy of mention while preparing the data for training the model is the challenge to deal best with the imbalanced medical data set as in this case. Therefore, to overcome this issue, future studies should focus on modern technical methods in dealing with imbalanced datasets to improve the DDI event prediction power without affecting the truth-ground medical data when the numbers of DDIs types are extremely imbalanced as reverse to the input source DrugBank. Furthermore, the latter versions of *RDKit* should be used to generate new set of chemical features. Using these versions, feature sets such as Morgan fingerprint would be integrated and considered to improve the predictive performance. Finally, with innovative methods (e.g., GCNN [29,30]), a given input molecule can be extracted to key features and functionality, or motifs; and these methods can replace SSP to generate a more useful feature set. When a bigger DDI set have been created, GCNN is able to identify potential molecular motif liabilities for a given molecular pair rather than the use of SSP [31].

3.5. Practical contribution

Our proposed EDN model can be applied in the discovery of DDI for better managing the description of drugs in multiple treatment strategies. Considering adverse drug events as unwanted occurrences that clinicians, with all possible efforts, should avoid while prescribing medications for patients, our EDN model's practical application in predicting DDIs events with surprisingly high accuracy can provide a robust platform for managing this problem. Our EDN model confirms the capability of the ensemble method in helping to minimize the generalization error in the aggregated model while still exploiting the synergic prediction power of different base models with various algorithms.

4. Conclusion

This study proposes a novel ensemble neural networks model for more accurate predictions of drug-drug interactions. The proposed model is trained with the benchmark data that make use of the structural similarity profiles of drugs for the prediction task. Our proposed model shows improved classification performance over the baseline study. Our model's high accuracy performance places it among the top list of those well-built pharmacovigilance-assisted tools that facilitate the detection of drug-drug interactions to support medical decisions and drug development.

CRediT authorship contribution statement

Thanh Hoa Vo: Conceptualization, Methodology, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing, Visualization. **Ngan Thi Kim Nguyen:** Validation, Writing – original draft. **Nguyen Quoc Khanh Le:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

We released our source codes and models at https://github.com/khanhlee/edn-ddi.

References

[1] Kim J, Parish AL. Polypharmacy and medication management in older adults. Nurs Clin North Am 2017;52(3):457–68.
[2] Kantor ED, Rehm CD, Haas JS, Chan AT, Giovannucci EL. Trends in prescription drug use among adults in the United States From 1999–2012. JAMA 2015;314(17):1818–31.
[3] Zheng WY, Richardson LC, Li L, Day RO, Westbrook JI, Baysari MT. Drug-drug interactions and their harmful effects in hospitalised patients: a systematic review and meta-analysis. Eur J Clin Pharmacol 2018;74:15–27.
[4] Raschetti R, Morgutti M, Menniti-Ippolito F, Belisari A, Rossignoli A, Longhini P, et al. Suspected adverse drug events requiring emergency department visits or hospital admissions. Eur J Clin Pharmacol. 1999;54(12):959–63.
[5] Budnitz DS, Pollock DA, Weidenbach KN, Mendelsohn AB, Schroeder TJ, Annest JL. National surveillance of emergency department visits for outpatient adverse drug events. JAMA 2006;296(15):1858–66.
[6] Knox C, Law V, Jewison T, Liu P, Ly S, Frolkis A, et al. DrugBank 3.0: a comprehensive resource for 'omics' research on drugs. Nucleic Acids Res. 2011. D1035-41.
[7] Wishart DS, Feunang YD, Guo AC, Lo EJ, Marcu A, Grant JR, et al. DrugBank 5.0: a major update to the DrugBank database for 2018. Nucleic Acids Res 2018;46(D1):D1074–82.
[8] Vo TH, Nguyen NTK, Kha QH, Le NQK. On the road to explainable AI in drug-drug interactions prediction: a systematic review. Comput Struct Biotechnol J 2022;20:2112–23.
[9] Vilar S, Uriarte E, Santana L, Tatonetti NP, Friedman C. Detection of drug-drug interactions by modeling interaction profile fingerprints. PLoS One 2013;8(3):e58321.
[10] Vilar S, Uriarte E, Santana L, Lorberbaum T, Hripcsak G, Friedman C, et al. Similarity-based modeling in large-scale prediction of drug–drug interactions. Nat Protoc 2014;9:2147–63.
[11] Cheng F, Zhao Z. Machine learning-based prediction of drug–drug interactions by integrating drug phenotypic, therapeutic, chemical, and genomic properties. J Am Med Inform Assoc 2014;21(e2):e278–86.
[12] Park K, Kim D, Ha S, Lee D. Predicting Pharmacodynamic Drug-Drug Interactions through Signaling Propagation Interference on Protein-Protein Interaction Networks. PLoS One 2015:e0140816.
[13] Deng Y, Xu X, Qiu Y, Xia J, Zhang W, Liu S. A multimodal deep learning framework for predicting drug-drug interaction events. Bioinformatics 2020;36(15):4316–22.
[14] Rohani N, Eslahchi C. Drug-Drug interaction predicting by neural network using integrated similarity. Sci Rep 2019:13645.
[15] Lee G, Park C, Ahn J. Novel deep learning model for more accurate prediction of drug-drug interaction effects. BMC Bioinf 2019;20(1):415.
[16] Shtar G, Rokach L, Shapira B. Detecting drug-drug interactions using artificial neural networks and classic graph similarity measures. PLoS One 2019;14(8):e0219796.
[17] Lee CY, Chen YP. Descriptive prediction of drug side-effects using a hybrid deep learning model. Int J Intell Syst 2021;6(6):2491–510.
[18] Ryu JY, Kim HU, Lee SY. Deep learning improves prediction of drug–drug and drug–food interactions. Proc Natl Acad Sci U S A 2018;115(18):E4304.
[19] Mahmud SMH, Chen W, Meng H, Jahan H, Liu Y, Hasan SMM. Prediction of drug-target interaction based on protein features using undersampling and feature selection techniques with boosting. Anal Biochem 2020;589:113507.

[20] Rogers DJ, Tanimoto TT. A computer program for classifying plants. Science 1960;132(3434):1115–8.
[21] Rogers D, Hahn M. Extended-Connectivity Fingerprints. J Chem Inf Model 2010;50 (5):742–54.
[22] Hung TNK, Le NQK, Le NH, Tuan LV, Nguyen TP, Thi C, et al. An AI-based prediction model for drug-drug interactions in osteoporosis and Paget's diseases from SMILES. Mol Inform 2022;41(6):2100264.
[23] Zhang W, Jing K, Huang F, Chen Y, Li B, Li J, et al. SFLLN: a sparse feature learning ensemble method with linear neighborhood regularization for predicting drug–drug interactions. Inform Sci 2019;497:189–201.
[24] Nyamabo AK, Yu H, Shi J-Y. SSI–DDI: substructure–substructure interactions for drug–drug interaction prediction. Brief Bioinform 2021;22(6).
[25] Lin S, Wang Y, Zhang L, Chu Y, Liu Y, Fang Y, et al. MDF-SA-DDI: predicting drug–drug interaction events based on multi-source drug fusion, multi-source feature fusion and transformer self-attention mechanism. Brief Bioinform 2021.
[26] Tng SS, Le NQK, Yeh HY, Chua MCH. Improved prediction model of protein lysine crotonylation sites using bidirectional recurrent neural networks. J Proteome Res 2022;21(1):265–73.
[27] Zhang W, Chen Y, Liu F, Luo F, Tian G, Li X. Predicting potential drug-drug interactions by integrating chemical, biological, phenotypic and network data. BMC Bioinf 2017;18.
[28] Ferdousi R, Safdari R, Omidi Y. Computational prediction of drug-drug interactions based on drugs functional similarities. J Biomed Inform 2017;70:54–64.
[29] Park C, Park J, Park S. AGCN: Attention-based graph convolutional networks for drug-drug interaction extraction. Expert Syst Appl 2020;159:113538.
[30] Wang F, Lei X, Liao B, Wu FX. Predicting drug–drug interactions by graph convolutional network with multi-kernel. Brief Bioinform 2021;23(1).
[31] Feng Y-H, Zhang S-W. Prediction of drug-drug interaction using an attention-based graph neural network on drug molecular graphs. Molecules 2022;27(9):3004.